

# Configuration-level split aliasing in object-detection benchmarks: how the reported split is selected on, how far the selection transfers, and what a three-way protocol reveals

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**Working title (v0.1). Target venue: Pattern Recognition (Elsevier).**

**Provenance of this file.** The English submission form of the P2 manuscript, a faithful

translation of the Chinese governing draft 评测有效性稿\_正文\_v0.1.md

(md5 fe880ca0e3af77983382065db75255f3, the state it was translated from, 2026-09-18; that draft is a live

document, re-hashed whenever it changes — see `work/verify_text_hashes_20260917.py`); every number is unchanged and backed by the

local artifacts listed in the Supplementary Material. Where the drafts disagree, the Chinese governs.

Citation numbers refer to P2\_参考文献\_v0.1.md (**40 entries, all verified**), seven added

in r39 ([34]–[40]) for the venue's 35–45 reference band, all verified against Crossref or DataCite

DOI records; this form cites **every** entry, including the benchmarks and frameworks that the

governing Chinese draft names without numbers. The four citations unresolved in the first

draft — [A] the CVPR 2026 geospatial leakage audit, [B] AI-TOD, [C] UAVDT,

[D] D-Fire — are **now verified** as [30] [31] [32] [33]; two of the four titles and

one venue had differed in an unverified citation (see that file, §4).

## Abstract

Object-detection papers report a number from a split they call held out. We audit the configuration

behind that premise: **6 of 14** YOLO configurations we produced bind `val:` and `test:` to one

directory, so the reported checkpoint was selected on the reported split. Training and evaluation do

not overlap at the image level — the intersection is exactly zero — so this is not contamination but **checkpoint**

selection<sup>\*\*</sup>, positive in all four cells measured on 569 runs (all four now powered; §5.1), the baseline

arm gaining more. Its

direction is **not single-valued**: in 19 public benchmarks the reported number comes from a

non-independent split in **12/19** (§6.1, single-rater: a second reading), yet four all-correct answers exist by counting unit. A three-way split

remediation reproduces the headline effect (smoke→SFCHD **+1.434 pp**, 10/10 seeds positive) while adding

seeds lowers the mean by **9 %–29 %**, a direct measure of seed inflation.

**Keywords:** evaluation validity; dataset splits; checkpoint selection; object detection; benchmark audit;

reproducibility

## 1. Introduction

A detection paper reports mAP50-95 on a split it calls held out, implying a number that estimates

performance on unseen data. The inference needs a premise the paper rarely states: the checkpoint was not

chosen on that same split. Binding `val:` and `test:` to one directory breaks it silently — the pipeline evaluates

there, keeps the best-scoring checkpoint and reports its score: a **maximum**, not a sample. Surveys of the detection literature [36] trace how that number became the

field's unit of comparison; this paper audits the reporting convention producing it.

That premise is audited in three places, in increasing distance from our own work: **our own corpora**,

where we count how many of the YOLO configurations we produced alias the two keys; <sup>\*\*19</sup> public

benchmarks, **audited row by row for which split the reported number comes from**; and 20 published

third-party training logs\*\*, where we measure the premium directly, in data we did not produce.

Three further things we have not seen done here. We **remediate** with a three-way split separating the

selection set from the reported set, plus a recomputable protocol, reporting the consequences, not only the

fix. We **measure the premium's fate**: best and last checkpoints and both splits exist, so the full

$2 \times 2$  grid is computable: how much of the val-selected premium is realized on an independent split? And we **pre-register** a generalization hypothesis, then report that it misses its criterion.

### 1.1 Why the unit matters

Ask "how many of these benchmarks have a split problem?" and, unless a unit is declared, every answer is

defensible and they disagree by a factor of six: 2/19 (the release artifact binds both keys to one path),

10/19 (no independent held-out test is provided), 13/19 (the official protocol does not let you evaluate an

independent test locally), 12/19 (the reported number comes from a non-independent split). We therefore

treat "declare the unit" as a contribution, not a caveat (§3, §6).

### 1.2 What we do not claim

We do not claim that aliasing is data contamination: the image-level intersection between the

training images and the held-out evaluation images is **zero**. We do not claim a single bias direction: our paired analysis finds \*\*two corpora

where val understates the effect and two where it overstates it\*\*, nine cells not significant (§5.1). We

do not claim the convention is framework-general (§6.5), nor that the benchmarks we audit are wrong: they

report what their configuration computes.

**Positioning (one sentence).** We claim neither to be first to audit split problems in detection benchmarks

nor to audit image-level leakage: both are taken. We occupy a different cell — **configuration-level**

aliasing, a direction that varies by corpus, and the requirement that a count declare its unit.**\*\***  
Every number points to a local artifact; every claim narrowed after external checking is

recorded as narrowed (§5.4).

**Scope.** All quantitative claims about selection behaviour come from **YOLO-lineage pipelines**,  
the only

family among the five frameworks we checked selecting the checkpoint by the validation metric  
by default

(§6.5) — stated at the claim, not only in a limitations section.

## 2. Contributions and related work

### 2.1 The closest work, and where this paper sits

The table was verified against the **full texts** of both 2026 studies (PDFs archived; extraction  
script

retained).

**Table S1** → Supplementary §S8.

None of these works is wrong — **they audit images, we audit configurations**. Image-level  
duplication

inflates the metric (both 2026 studies measure that); configuration-level aliasing shifts the  
**\*\*comparison**

between arms, **whose direction we measure as** not single-valued**\*\***.

They instantiate the taxonomy of [34] and [35]: leakage is the mismatch between the data used to  
choose a

model and the data used to score it, duplication one such mismatch [3].

Binding `val:` and `test:` to one directory creates the same mismatch **without duplicating an  
image**, so

de-duplication cannot find it — why the two literatures have not met.

**Why this is a selection question, not only a leakage question.** The bias we measure is

the one the model-selection literature has described for decades: **\*\*choosing a model by its**

score on a validation set makes the reported score optimistic\*\* [41], [42], and the classical remedy is to *nest* the selection inside the evaluation rather than to de-duplicate the data.

In detection benchmarks that remedy is unavailable in practice \*\*until the configuration is read\*\*: the selection step is implicit — the training script writes `best.pt` by fitness and the paper reports that checkpoint — so a reader who trusts the split names sees a held-out evaluation where the pipeline in fact performed a **maximum**. The two literatures that do confront benchmark erosion — reused test sets [43], standard splits [44] — therefore describe a bias that this configuration reaches without any data being reused or mis-split.

We take [41], [42] as the general statement and report the detection-specific instantiation:

**the selection premium (best – last), and how much of it transfers to a disjoint split**

(§5, §8).

**The sharpest contrast** is *Drones* 2026 [2]: it too reports a val-vs-test gap, but concludes "no systematic direction", treating split agreement as evidence that model selection has not overfit validation. We measure a **different quantity**, in two tiers: (i) the selection premium (best – last) is positive in all four cells, the baseline arm gaining more (four paired CIs excluding zero); (ii) the paired  $\Delta$ gap on three-way corpora **varies in sign** (2 cells where val understates, 2 where it overstates, 9 not significant). We therefore do **not** write "understated" unqualified, and state the difference from [2]: [2] reports a

raw val-minus-test gap, mixing two splits' difficulty, **not identifiable**, whereas our \*\*paired difference\*\* cancels difficulty and still shows a residual shift. [30] is supporting, not competing: cited only as evidence that the genre has a

venue, not as a comparable quantity.

## 2.2 Contributions

1. **We quantify the magnitude and direction of configuration-level aliasing.** The selection premium

(best – last) is **positive in all four cells, the baseline arm gaining more** (four paired 95 % CIs excluding zero, §5.1 Tier 1). **All four cells are powered (n = 10–11)** — the fourth,

dota15, entered at  $n = 3$  and was escalated to  $n = 10$  under a criterion fixed in advance by the shared review protocol (Supplementary S6) before

the runs existed; it is met (§5.1)  $\Rightarrow$  the convention systematically shifts **arm-to-arm comparisons**; yet

the shift's **sign varies by corpus** (three-way paired differencing: 2 cells where val understates, 2 where it overstates, 9

not significant — and the four crossing cells are now **all at  $n = 10$** , the two overstating ones only after the escalation in §5.1).

**We do not claim a general understatement** (§5). It appears in third-party logs too (20 logs / 6 sources, median +1.04 pp, max +5.65 pp,

18/20 positive; §5.5) — a sample, not a field-level rate.

*Note: image-level\** leakage magnitude is quantified by the two 2026 studies;

**configuration-level** aliasing magnitude has not been measured before.\*

2. **We show that "how many benchmarks have a problem" needs a declared unit**, and give a recomputable

table in four units with per-row evidence classes (§3, §6): four defensible answers.

3. **We give a remediation protocol and report its consequences**. Under a three-way split the effect

reproduces ( $n = 10$ ), while adding seeds lowers the mean by 9 %–29 % (§8) — a direct measurement of

selection inflation; the premium's realization becomes measurable (**19 %**, §8.6).

4. **We disclose our own defects** rather than removing them: our analysis' first version had non-reproducible items — scripts depending on a rented pod, a citation to a file not containing the quoted numbers, no written endpoint convention. §10, not deleted.

### 2.3 What we explicitly do **\*\*not\*\*** claim (shared tooling)

This paper and the budgeted fine-tuning study it re-measures — the companion paper — share **\*\*process**

**protocols\*\***: a multi-model review checklist and a local/cloud workflow discipline (scripted remote

execution, all-or-nothing patching, audit anchors, a local artifact pointer behind every number).

**\*\*Both** are project-internal protocols, not publications, and not a

methodological contribution of this paper.\*\* By agreement, neither paper may present them as its own

methodology. This paper displays only their **outputs**: the six self-reported defects in §10 and the artifact pointer behind every number. So does the **audit-anchor system** (a mechanically checked set of anchors that must all

pass on every rebuild of this manuscript): shared tooling, no contribution claimed. Whether those documents appear in a

submission is a joint decision of the two papers; this paper asserts only that the same protocols were

used, and originality is not claimed. Corresponding statements appear in the companion.

### 3. Units: the methodological premise

"Have a split problem" is a property of a (**benchmark, counting unit**) pair, not of a benchmark. We use four units, defined by **who controls the split**:

Table S2 → Supplementary §S8.

**The `reported` unit is read from the release documentation and from the dataset's own paper:** we did not sample papers per benchmark and make no claim about how often the community reports a given split; a stated-frame literature sample is future work (§12). For each of 19 benchmarks we record all four units, an evidence class and a confidence level, and recompute every percentage from the row markers under a per-unit subtotal check. The full  $19 \times 4$  table, the subtotal checks and the cross-check against the original audit are **Supplementary S1**; the derived counts are in §6.1.

**The premise we rely on throughout:** a "yes" under any unit means *the reported number is not a sample from an independent split*. It does **not** mean the benchmark is broken or the experiment dishonest; the number does not license a generalization claim.

## 4. Finding 1: our own corpora

### 4.1 How common the aliasing is

In **our own** experimental corpora, **6 of 14** bind `val:` and `test:` to the same directory. Counting by file, not by corpus, the whole-tree scan covers **19 YAML files**; on our own manifest the ratio is **8 of 28**. One case is maximal: for **SHWD all three keys** [25] (`train:`, `val:`, `test:`) **point to a single directory**.

**This is not contamination.** The image-level intersection is **exactly zero**: training and evaluation images do not overlap. It changes not *what* was evaluated but *which checkpoint* was reported, turning "selection contamination" from an ethical question into a **measurement** question.

## 4.2 The failure is produced by a tool, not by carelessness

A PASCAL VOC [13]→YOLO conversion script wrote two keys from one `images_dir`, so `train == val` appears **because the generator could not produce distinct values** (§6.1 records the same for Mendeley face-mask [27]). §11's first recommendation: make "are val and test the same source?" a **pipeline-level hard check** — human YAML review will not catch it.

## 5. Finding 2: the selection premium — magnitude, direction, and mechanism

**Evidence.** `eval_validity/selection_premium_local_20260916.txt`, reading **569 published runs** from the local archive (the local version is the 2026-09-16 repair).

**Definition.** `premium` = **best-epoch – final-epoch** mAP50-95 **within a single run**. When `val == test`, "best" was selected on the reported split, while `last` and `last5` are endpoints **no selection rule can reach**. This is the checkpoint-selection term of the run-to-run variation studied in [37] (early stopping, data order) and [38] (benchmark variance across seeds): what those works average over as uncertainty, we measure as a reproducible component.

Table S3 → Supplementary §S8.

### 5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus

The direction claim **must be reported in two tiers**: collapsing them into "the convention understates the effect" is wrong — external checking refuted it (§5.4, II).

**Tier 1: the selection premium is positive in all four cells, and the baseline arm gains more.**

The tested quantity is the paired difference  $d(\text{seed}) = \text{premium\_baseline}(\text{seed}) - \text{premium\_strategy}(\text{seed})$ , within cell and seed:

**Table 1.** Tier 1 — the selection premium per arm and the paired difference between arms, by cell. Source: `verify_two_critiques_20260916.txt`.

| Cell                      | n   | premium baseline | premium strategy | paired diff   | sd    | t    | p                    | 95 % CI          |
|---------------------------|-----|------------------|------------------|---------------|-------|------|----------------------|------------------|
| ---                       | --- | ---              | ---              | ---           | ---   | ---  | ---                  | ---              |
| smoke2sf (strong)         | 11  | +0.799           | +0.382           | <b>+0.416</b> | 0.240 | 5.74 | $1.9 \times 10^{-4}$ | [+0.255, +0.578] |
| shwd2sf (corroborating)   | 11  | +1.038           | +0.512           | <b>+0.526</b> | 0.207 | 8.41 | $7.6 \times 10^{-6}$ | [+0.386, +0.665] |
| a2d15 (directional probe) | 10  | +0.534           | +0.262           | <b>+0.272</b> | 0.183 | 4.69 | $1.1 \times 10^{-3}$ | [+0.141, +0.403] |
| dota15 (within-domain)    | 10  | +0.941           | +0.492           | <b>+0.449</b> | 0.196 | 7.23 | $4.9 \times 10^{-5}$ | [+0.309, +0.589] |

**Table 2.** Tier 2 — the paired difference  $\Delta\text{gap}$  by corpus, with the reading it supports.  $\Delta\text{gap} > 0$  means val understates the strategy arm's effect. Source: xeval\_analysis\_20260916.txt.

| Corpus        | n    | $\Delta\text{gap}$ (pp) | p                    | 95 % CI          | Reading         |
|---------------|------|-------------------------|----------------------|------------------|-----------------|
| ---           | ---  | ---                     | ---                  | ---              | ---             |
| shwd2sf       | 10   | <b>+0.779</b>           | 0.0053               | [+0.297, +1.261] | val understates |
| smoke2sf      | 10   | <b>+0.461</b>           | 0.0263               | [+0.068, +0.855] | val understates |
| p_aitovis     | 10   | <b>-1.226</b>           | $1.3 \times 10^{-8}$ | [-1.370, -1.081] | val overstates  |
| p_vistod15    | 10   | <b>-3.315</b>           | $6.5 \times 10^{-9}$ | [-3.676, -2.953] | val overstates  |
| other 9 cells | 3–10 | -2.46 to +13.66         | $\geq 0.054$         | all include 0    | not significant |

(The third §5 evidence table, S5, is in Supplementary §S8.)

**The four paired differences are +0.416 / +0.526 / +0.272 / +0.449 pp** (Table 1, in cell order) — **all four cells powered at  $n = 10\text{--}11$ , and all four 95 % CIs excluding zero** (p from  $7.6 \times 10^{-6}$  to  $1.1 \times 10^{-3}$ ). dota15 entered this table at  $n = 3$ , where §7.1 permits only "significant under the paired t-test; the non-parametric test is unavailable at this  $n$ ", and that three-run cell included two runs without a seed identity (§5.3, §10). Both are now history rather than a caveat: the escalation below supplies ten seeded runs and a permutation-valid p. The right scale is the SE of the difference (**0.058–0.120 pp**): \*\*each arm is a mean over

10–11 seeds. **Runs are independently randomized\*\***, so pairing brings

**no variance benefit**: the paired t of 5.74 and the unpaired (Welch) t [29] of **5.68** agree — **the conclusion does not depend on pairing**. (The critic's 0.515 pp is the **single-run**

data-order component of §8.4; see §5.4 (I).) Endpoints no selection rule can reach make the paired contrast **larger**:

Paired t-tests are all  $p \leq 1.4 \times 10^{-9}$  ( $n = 10\text{--}11$ ). **Tier 1 may be stated as directional**: the baseline arm

benefits more from selecting on the reported split.

**The  $n = 3$  cell was escalated, and the criterion fixed for it in advance was met**. Tier 1's

fourth cell, dota15, entered this table at  $n = 3$  with a parametric  $p = 0.039$ , below the resolution of any sign-flip test there (§7.1); the pre-declared consequence was that a ten-seed paired difference

failing to be positive with  $\geq 8/10$  same-sign and a CI excluding zero would remove the cell from the

four-cell headline. Ten seeds (42–51) give **+0.449 pp**,  $t = 7.23$ ,  $p = 4.9 \times 10^{-5}$ ,

95 % CI [**+0.309, +0.589**], 10+/0— **the criterion is met as stated**, so the cell is read as

powered rather than directional. Its mean is **24 % below** the  $n = 3$  value ( $+0.589 \rightarrow +0.449$ ), a

fourth instance of §8.4's seed inflation inside the published 9 %–29 % band. Per-seed values, the

recipe-identity check across the cell's twenty `args.yaml`1, and the GPU-memory sensitivity run are in

### Supplementary S28.

**Tier 2: on the same three-way corpora, the paired difference  $\Delta_{\text{gap}}$  varies in sign.** The raw quantity

$M_{\text{val}}(\hat{e}) - M_{\text{test}}(\hat{e})$  is **not identifiable**: it ranges from **–33.0 to +21.2 pp**, and with

**no per-epoch test evaluation** the components cannot be separated. What remains is the **\*\*paired difference\*\***, each arm's gap subtracted within seed, which cancels the difficulty term:

$\Delta_{\text{gap}} = (\text{arm difference on val}) - (\text{arm difference on test})$ ;  **$\Delta_{\text{gap}} > 0$  means `val` understates the strategy arm's effect**

The four cells crossing the line: **+0.779 pp** (shwd2sf) and **+0.461 pp** (smoke2sf) — **val**

**understates**; **–1.226 pp** (p\_aitovis) and **–3.315 pp** (p\_vistod15) — **it overstates**.

**All four are now powered and none is  $n = 3$ :** the two overstating cells were escalated from  $n = 3$

to  $n = 10$  under criteria fixed in advance, and both met them (reported immediately below).

Four of 13 cells cross the uncorrected line at

$p < 0.05$ ; three marginal cells (aitod20 +1.481,

$p = 0.054$ ; fire +13.658,  $p = 0.061$ ; mende20 +2.472,  $p = 0.099$ ) **do not cross the line and we make no claim about them**. **\*\***These 13 cell tests are reported

uncorrected, as a descriptive scan. Across the 13, a Benjamini–Hochberg correction at  $q = 0.05$

**now leaves three cells significant** — p\_vistod15 ( $6.5 \times 10^{-9}$ ) and p\_aitovis ( $1.3 \times 10^{-8}$ ) far

below their lines, and shwd2sf (0.0053 against a line of 0.0115), because the two very small

p-values at the front of the ordering lift the lines behind them; smoke2sf (0.0263 against

0.0154) does not survive. **This reverses what we reported before the escalation**, when the smallest p (0.0053) lay above the first line (0.00385) and nothing survived — the change comes from

measuring two cells at  $n = 10$  instead of  $n = 3$ , not from choosing a different test. So the Tier 2 sign pattern now also carries three cells that survive the paper's own family correction; §9 applies BH within its own family of three registered pairs, where the count is stated separately. Proposition 5 of §13 gives the formal reason this count depends on the declared family.\*\*

**The two overstating cells were escalated too, and both met their pre-declared criteria.** The plan fixed before the runs (40 runs, seeds  $42-51 \times 2$  arms  $\times 2$  cells; paired *and* sign-flip tests; {best, last}  $\times$  {val, test}; a prespecified correction) named the strong branch of its outcome rule:

*"both remaining negative and significant with permutation  $p \leq 0.002$ ".* At  $n = 10$ :  $p_{\text{aitovis}}$

**-1.226 pp** (95 % CI [-1.370, -1.081],  $t = -19.17$ ,  $p = 1.3 \times 10^{-8}$ , **0+/10-**, permutation

$p = \mathbf{0.001953}$ ) and  $p_{\text{vistod15}}$  **-3.315 pp** ([-3.676, -2.953],  $t = -20.76$ ,  $p = 6.5 \times 10^{-9}$ ,

**0+/10-**, permutation  $p = \mathbf{0.001953}$  —  $2/2^{10}$ , the attainable floor at  $n = 10$ ). \*\*Both conditions

are met, **including** Holm across the two targets, **so** the abstract stays as written and its weak

form is strengthened\*\*: "val understates in two cells and overstates in two" no longer rests on two

$n = 3$  rows. The endpoint the comparison rests on was identified rather than assumed — our

re-evaluation of best.pt on test reproduces the archived `sio_b_results.csv` to 0.0046 pp — and

the per-seed values, the endpoint check and the auxiliary quantities are in **Supplementary S29**.

**The sharpest single fact:** in the shwd2sf cell the arms differ

on **val** by  $60.669 - 60.397 = \mathbf{+0.272 \text{ pp}}$  (baseline better) and on **test** by  $43.559 - 44.066 =$

**-0.507 pp** (baseline worse). **The sign reverses**, so reading only val inverts the arm ordering.

**The paper's formal claim is therefore:** the convention systematically shifts **arm-to-arm comparisons**

(Tier 1: four cells, same sign, CIs excluding zero); **the direction of that shift varies by corpus**

(Tier 2: 2 understate / 2 overstate / 9 not significant) — \*\*neither "a general understatement" nor

"no systematic direction".\*\*

**Scope (cross-framework boundary).** Every piece of evidence here comes from **\*\*YOLO-lineage**

**pipelines\*\***. We checked five

mainstream frameworks at pinned commits: **only the YOLO lineage does this by default** — Detectron2

(TEST.EVAL\_PERIOD = 0) [7], MMDetection [6], DETR [10]

and PaddleDetection [9] **report the final epoch instead** (§6.5). **\*\*"The convention"** here

therefore means that pipeline family's convention, not the field's.\*\*

**Feasible cross-validation.**  $M_{\text{test}}(\hat{\epsilon})$  arm differences from `sio_b_results.csv` — smoke2sf

+1.434, shwd2sf +0.507 — match the published clean-protocol readings of §8.2 **at the printed precision**:

independent evidence that two machines and protocols agree.

## 5.2 Two versions must be disclosed side by side

The same script on the **pod root** (347 runs,  $n = 10$ ) gives a smoke2sf premium of **+0.786 / +0.390**;

on the **local archive** (569 runs,  $n = 11$ ) **+0.799 / +0.382** — agreeing in sign and magnitude,

not value. **The main text uses the local version**; the pod version is recorded as a cross-check.

## 5.3 The dota15 cell: now reproducible, and one wording defect exposed

**This cell was not reproducible from the release as first written, and the repair exposed a wording defect** (both disclosed, §10 row 2). The repair, the value-by-value agreement and the defect are recorded in **Supplementary S12**.

## 5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

**Two external critiques were checked: one refuted, one judged non-identifiable.** The refuted one compared the premium difference against a single-run component (0.515 pp) rather than the standard error of a difference of arm means; the other asked for the raw  $M_{\text{val}} - M_{\text{test}}$ , whose difficulty term cannot be separated from selection optimism. **We used to write** that the gap was simply unidentifiable in this archive; the external check **corrects this to** the paired-difference form, which cancels the difficulty term under the pairing assumption stated in §13. Details and the recomputation are in **Supplementary S12**.

## 5.5 The same component in third-party public logs (independent evidence)

**The same quantity appears in other people's logs**: 20 logs from 6 sources, median **+1.04 pp**, max **+5.65 pp**, 18/20 positive — a sample, not a field-level rate. **The sampling frame**: logs found by repository **search** (Kaggle and public repositories), with the exact queries in

**Supplementary S12**; it is a sample, and public logs **cannot establish aliasing** at the configuration level. The rest of the frame and what the sample cannot support are in **Supplementary S12**.

## 5.6 Why $\Delta_{\text{gap}}$ 's sign varies by corpus: three of five candidate mechanisms are refuted by the data

**Three of five candidate mechanisms are refuted and the mechanism itself is not identifiable:** the sign of  $\Delta_{\text{gap}}$  is a sum of same-order terms, so no single mechanism can be read off it (§13, Proposition 3). **The paired quantity is nonetheless identifiable and is what we report; the raw  $\text{M}_{\text{val}} - \text{M}_{\text{test}}$  is not identifiable at all.** The candidate-by-candidate examination, the per-cell components and the exact identity are in **Supplementary S12** and Table S8–

S10.

## 6. Finding 3: how general this is across public benchmarks

**Evidence:** `benchmark_split_audit.md`, a line-by-line audit of 19 benchmark rows, plus

`split_units_19rows_20260916.md` (recomputation script; no hard-coded percentages).

The 19 audited benchmarks, with the release or dataset paper read for each: COCO [18], PASCAL VOC [13], Objects365 [14],

Open Images v7 [15], DOTA v1.0 and v2.0 [22], VisDrone-DET [23], AI-TOD [31], UAVDT [32], xView [19],

DIOR [21], NWPU VHR-10 [16], SHWD [25], SFCHD [26], MAFA [24], Mendeley face-mask [27], WIDER FACE [17],

CrowdHuman [20], D-Fire [33]. The question is **evaluation validity**. Dataset-level bias auditing [39]

and classifier-accuracy estimation [40] are neighbours: both ask whether a reported number measures what it

is taken to measure, but neither audits a release's configuration binding.

### 6.1 The result under four units (single-rater judgements)

**Table 3.** The four counting units over the 19 audited benchmarks. **A count is meaningless without its unit** (§3). Source: `split_units_19rows_20260916.md`.

| Unit                                             | non-independent                        | of decidable rows                     | undecidable |
|--------------------------------------------------|----------------------------------------|---------------------------------------|-------------|
| ---                                              | ---                                    | ---                                   | ---         |
| <b>release</b> (the benchmark's own release)     | <b>10/19 <math>\approx</math> 53 %</b> | 10/19                                 | —           |
| <b>protocol</b> (the official protocol)          | <b>13/19 <math>\approx</math> 68 %</b> | 13/19                                 | —           |
| <b>yolo_dist</b> (the generic YOLO distribution) | <b>4/19 <math>\approx</math> 21 %</b>  | <b>4/15 <math>\approx</math> 27 %</b> | 4 rows      |

|                                                                                      |                                        |                     |        |
|--------------------------------------------------------------------------------------|----------------------------------------|---------------------|--------|
| package)                                                                             |                                        |                     |        |
| <b>reported</b> (the layer actually reported)                                        | <b>12/19 <math>\approx</math> 63 %</b> | <b>12/15 = 80 %</b> | 4 rows |
| ↳ of which <b>literal two-key same path</b> (= the 2/19 the original audit reported) | <b>2/19 <math>\approx</math> 11 %</b>  | 2/19                | —      |
| <b>reported – clean</b>                                                              | <b>3/19 <math>\approx</math> 16 %</b>  | 3/15                | 4 rows |

**These 19 rows were coded by one rater** (§10 defect 5 records that the audit is hand-written, with

28 literal judgements and no second rater), so every count below is a single-rater count.

**Which number the `reported` unit judges.** For one row the answer depends on which reported number is meant: COCO. Community tables report the **server-scored `test-dev`** value, which no local run can be selected on, and we count that as independent; a paper's own ablations report its **`val`** value, which is also the selection split. On the first reading reported is **12/19** and **3/19** are clean; on the second, **13/19** and **2/19**. No other row is affected. This article prints the first reading in the abstract and states both here.

**Subtotal checks:** reported **12 + 3 + 4 = 19**; yolo\_dist **\*\*4 + 5** clean + 6 with no test: key + 4 undecidable = **19\*\***; release **10 + 9 = 19**; protocol **13 + 6 = 19**.

**Correction to the first version of this table.** It **\*\*put 2/19** under

yolo\_dist, **contradicting the recomputation: `yolo\_dist` is 4/19 (PASCAL VOC and SFCHD's literal**

two-key alias, SHWD's **`train == val`**, Mendeley's **generator-produced alias**), while 2/19 is only a subset

of it — **the A-evidence, literal `val == test` rows the audit reported. The table makes a unit a whole column\*\***, ordered as §3's four definitions.

**Algorithm self-check:** on the **literal two-key subset** the table **reproduces the audit's 2/19** (PASCAL VOC, SFCHD), so the difference comes from **the unit**, not from a transcription error; the whole yolo\_dist column is **4/19**.

**One row belongs to a different unit.** Mendeley face-mask's **train == val** comes from a **generator**

(a VOC→YOLO conversion script writing two keys from one images\_dir), not a distribution package, so it is

not counted in the 2/19 and must be described separately.

**A number we deliberately do not use.** The audit's "63–79 %" upper bound (15/19) has **\*\*no enumeration\*\***; we do not cite it.

**6.2 "Same directory" has four levels, and the audit reported only the narrowest**  
**"Same directory" has four levels, and the original audit reported only the narrowest.** The levels, the rows each admits, and the resulting shift in the headline number are in **Supplementary S12** and Table S12.

**6.3 We do **\*\*not\*\*** reproduce the original audit's two numbers, and we say why**  
**We do not reproduce the original audit's two numbers, and we say why:** one rests on the narrowest reading of "same directory", the other on a stale snapshot. The recomputation and the reason for each difference are in **Supplementary S12** and Table S13.

**6.4 One judgment the evidence does not support (corrected)**

The audit classified the FireSmoke mirror family row as "mixed" solely from a config with

`train == val`, but **another line of the same document records that this config is based on SHWD**. All

three FireSmoke/D-Fire copies have **three separated keys**, so D-Fire moves to the clean group.

**6.5 Across frameworks: is the convention specific to YOLO?**

**Only the YOLO lineage does this by default.** At pinned commits, Detectron2 (`TEST.EVAL_PERIOD = 0`), MMDetection, DETR and PaddleDetection **report the final epoch instead**; the file-by-file evidence (23 official configuration files) is in **Supplementary S12** and Table S14. **This is a targeted sample at pinned commits, not a census** of the five frameworks, and "the convention" therefore means that pipeline family's convention, not the field's.

## **7. Finding 4: what "significant" means, and how the endpoint convention changes it**

**7.1 With  $n = 3$  a permutation test **\*\*cannot\*\*** reject in principle**

**For a sign-flip permutation test on  $n$  pairs, the smallest attainable two-sided  $p$  is  $2/2^n$  — for  $n = 3$ ,**

**0.25.** Every  $n = 3$  cell still in our tables has a permutation  $p$  of 0.25: **not** "not significant"

but "no resolution at this  $n$ ". **\*\*After the two escalations reported in §5.1, no cell of the headline tables (Tables 1–2) is  $n = 3$  any more\*\***; the  $n = 3$  cells that remain are the nine non-crossing rows

of the 13-cell scan in Supplementary Table S8.

**Consequence.** An  $n = 3$  cell may only read "\*\*\*significant under the paired t-test; the non-parametric

test is unavailable at this  $n$ ", not\*\* a bare " $p = 0.025$ " — one reviewer question, "which test is that?", exposes the difference.

The attainable floors are:  $n = 3 \rightarrow 0.25$ ;  $n = 6 \rightarrow 0.031$ ;  $n = 7 \rightarrow 0.0156$ ;  $n = 10 \rightarrow 0.0020$ .

## 7.2 Both conventions become significant only at $n \geq 6$

**Both conventions become significant only at  $n \geq 6$**  — below that a permutation test has no resolution (§7.1). The clean-protocol seed extension that reaches  $n = 10$ , and the per-cell numbers, are in **Supplementary S12** and Table S15.

**First-report attribution (a convention we hold to).** The two  $\Delta$  values here (+1.434 / +0.507 pp) are the **primary readings of the companion paper**, our budgeted fine-tuning study: this paper cites them, marks them as such, and does **not** claim them as first reports. What this paper reports first is the three-way protocol, the selection premium (§5), and its consequences (§8.3). The full statement — including how the companion is named to the editor while it carries no identifier — is in **Supplementary S12**.

## 7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)

**One run is recorded with two different endpoints in two archives** — "final" (67.12) in one and "best" (70.06) in the other — i.e. **which checkpoint is recorded was never specified**; §10 discloses it and the two readings' consequences (significant versus not) are worked out in **Supplementary S12** and Table

S16.

# 8. Finding 5: a remediation protocol and its consequences

## 8.1 The protocol

**The protocol:** three-way corpora, YOLOv12n [11] + shapeiou, a 20 %-budget fine-tune at 100 epochs, seeds 42–51. The steps, the carve construction, the run inventory and the disjointness audit are in **Supplementary S12**.

## 8.2 Results: the effect reproduces, but the magnitude falls

Table S17  $\rightarrow$  Supplementary §S8.

**Both readings must be stated:** the effect **reproduces** under a protocol separating selection from reporting, and the magnitude is not comparable with the old protocol (different unit). All

reported metrics come from the **test** split (`sio_b_results.csv`), **not** the `per-epoch_results.csv` values — a different quantity; our first version **took the wrong column** and the fix is on the record.

**A defect in the document that reports these numbers.** The clean-protocol results document **\*\*cites a**

ledger file that is not merely misnamed<sup>\*\*</sup>: neither `_r10_final_stats.txt` nor the file later identified as

the "real log" (`_r10_stats_0403.txt`) contains **any** of that document's headline values (6 probes each,

0/6 hits)  $\Rightarrow$  **these numbers have no surviving stdout ledger.** We **do not** point at a replacement file

(renaming would only manufacture a new false citation). Where this paper cites those numbers it says so;

the p-values can be recomputed from `pvalue_table_local.py`.

**Recomputation status (updated 2026-09-16).** These readings were originally **not locally recomputable**:

the local archive lacks 6 of 7 corpora, which the script that *constructs* the three-way split needs (defect 5 in §10). We therefore ran a  **$2 \times 2$  evaluation** on a GPU machine — 41 three-way runs  $\times$   $\{\text{best, last}\} \times \{\text{val, test}\}$ , **164 evaluations, 0 failures** — and pulled back the per-epoch curves, `sio_b_results.csv` and the matrix: **84 files with a SHA-256 list** (`xeval_20260916/`).

**This section's readings and §5.6's decomposition are therefore locally recomputable;**  
**\*\*retraining is**

**not\*\***, needing the corpora. The batch also reproduces the archived `best  $\times$  test` values to within **0.0047 pp** (median 0.0021 pp) across all 41 runs.

### 8.3 A direct measurement of selection inflation

**A direct measurement of seed inflation:** adding seeds lowers the mean by **9 %–29 %** of the headline effect; the estimand is the change in the mean, not a single-run quantity. The design, the per-cell changes and the check against recomputed means are in **Supplementary S12** and Table S18. **The section title is deliberately the weaker, sample-scoped claim** ("the excluded one is the smaller in this sample"), since with three initializations the interval is wide: the components are **consistent with, not established by**, these three initializations.

## 8.4 The three variance components: the excluded one is the smaller in this sample

**The three variance components, and the excluded one is the smaller in this sample:** the single-run data-order component is **0.515 pp**, against initialization and seed components of the same order. What was held fixed, and why this is a sample statement rather than a general one, is in **Supplementary S12** and Table S19.

## 8.5 Why "same-pipeline pairing" is a necessity, not a habit

**Same-pipeline pairing is a necessity, not a habit:** mixing two pipelines' val/test conventions would confound the split convention with the framework. The reasoning and the counterexample are in **Supplementary S12**.

## 8.6 The premium's realization rate: **\*\*19 %\*\***

**Measured on the GPU machine for this revision** (script `work/rcmd/xeval_3way.py`;

artifacts `xeval_20260916/matrix.csv`, per-run archives).

Separating the two raises a question unaskable

before: **how much of the gain selected on `val` survives on `test`?**

**Method:** the  $2 \times 2$  evaluation described above, with `split=test`, `batch=32`,

`imgsz=640`, `plots=False`.

**Table 4.** The premium selected on `val` versus what that same selection action realizes on the disjoint `test` split, per arm-cell. Source: `xeval_20260916/matrix.csv`.

| Cell     | Arm      | n   | prem_val | `prem_test`   | realization rate |
|----------|----------|-----|----------|---------------|------------------|
| ---      | ---      | --- | ---      | ---           | ---              |
| shwd2sf  | baseline | 10  | +1.469   | <b>+0.735</b> | <b>50 %</b>      |
| shwd2sf  | strategy | 10  | +0.865   | <b>+0.151</b> | <b>18 %</b>      |
| smoke2sf | baseline | 10  | +0.691   | <b>+0.507</b> | <b>73 %</b>      |
| smoke2sf | strategy | 10  | +0.605   | <b>-0.158</b> | <b>-26 %</b>     |
| aitod20  | baseline | 1   | +0.786   | <b>-0.377</b> | <b>-48 %</b>     |

**Two more cells, measured the same way — and they land at the other end.** The escalation in §5.1

adds four arm-cells to this measurement: realization rates **82 % / 104 %** (`p_aitovis`, baseline / strategy) and **56 % / 29 %** (`p_vistod15`), pooled **55 %** against the **19 %** above. We do

**not** re-average that headline — it is defined over §8.6's own four arm-cells and Table 4 stays as printed — but the contrast supports this section's own conclusion (§8.7): **\*\*the realization rate is not a constant across corpora\*, so "19 %" is a property of those\* cells rather than of the**

convention. (In `p_aitovis`'s strategy arm the realized premium slightly exceeds the selected one, 104 %, i.e. within noise on quantities of  $\sim 0.27$  pp.) Source: `work/x1_aux_20260918.py`.

**A theory-driven check on this very quantity.** §13 derives the realization rate as

$1 - \text{WC} / \text{premium}$ , where WC is the winner's-curse part of the selected gain — the piece that came

from the `val` split's own noise rather than from real progress. Two consequences are testable on the

archive, and both hold. ① The inequality is **per run**, not just on average: over the 81 runs whose two sides are both measured,  $\text{WC} = \text{prem\_val} - \text{prem\_test}$  averages **+0.463 pp** ( $t = 8.34$ ,

$p = 1.8 \times 10^{-12}$ ) and the realized premium is smaller in **68 of 81** runs. ② The rate should **\*\*fall as**

the `val` curve gets noisier\*\*: sorting those runs by the noise scale  $\hat{\sigma}$  of their own `val` curve

gives terciles of **65.6 % / 49.8 % / 13.8 %**, Spearman  $\rho = -0.466$  ( $p = 1.3 \times 10^{-5}$ ,  $n = 80$ ).

**\*\*This section's 19 % and §8.7's per-cell variation are therefore the same statement at different**

**noise levels, which is why we do not present 19 % as a constant of the convention. What does not\*\***

hold is the stronger form we first wrote down — that the premium itself scales with the noise as

$\sigma \sqrt{2 \ln E}$ : the regression slope is  $0.026 \pm 0.222$  ( $p = 0.82$ ,  $n = 101$ ), because the premium's *level*

is set by the shape of the `val` curve (how far the peak sits before the final epoch). §13 records

that withdrawal. Scripts: `work/theory_p4_test_20260919.py`,  
`work/theory_meta_and_tests_20260919.py`.

where  $\text{prem\_val} = V_{\text{max}} - V_{\text{final}}$  (**selecting on val**, the book value) and

$\text{prem\_test} = T_{\text{best}} - T_{\text{final}}$  (**realized on test**).

**Aggregate: `prem\_val` mean +0.883 pp  $\rightarrow$  `prem\_test` mean +0.172 pp — a realization rate of 19 %**

**(a run-level bootstrap puts the run-weighted figure at 32.7 %, 95 % CI [17.2 %, 47.9 %],  $P(\text{rate} \leq 0) = 0.000$ , so the interval excludes no-realization; script ``work/x2b_kappa_arms_20260918.py``).**

**The weighting is declared here:** the ratio of **\*\*unweighted arm-cell**

**means, each of the five arm-cells counting once, the single-run `aitod20` cell carrying 20 %\*\***

of the weight; two other defensible weightings give **32.3 %** (by runs,  $n = 41$ ) and **34.0 %**

(the four  $n = 10$  arm-cells only), and that most of the premium selected on val

is **not** realized on an independent split holds under all three. In

**two of the five arm-cells the realized premium is negative**: the selection action

does not merely fail to pay off, it **costs**.

(best.pt  $\times$  val versus the per-epoch maximum in

results.csv differs by up to 0.083 pp, since best.pt is selected by **fitness**, not necessarily by

mAP50-95 — itself a convention.)

**Four limitations:** ① the two cells escalated in §5.1 (p\_aitovis, p\_vistod15) now also reach

$n = 10$  and their runs and test-side evaluations are **local** (x1\_tier2\_20260918/), but the

remaining 9 cells' runs are **not local**; ② prem\_test uses only **best and last**, not a

per-epoch curve;

③ last.pt's val recomputation differs from the final results.csv row by  $\sim 0.02$  pp; ④ this section

does **not** touch the 19-benchmark census of §6.

## 8.7 The per-epoch view: the realization rate is not a constant

The runs of §8.6 **retain intermediate checkpoints** (save\_period=5: epoch0,5,...,95, **20 files**; **3,854**

epoch\*.pt). Our first version sampled every fourth checkpoint (**41 runs  $\times$  {0, 20, 40, 60, 80} = 202 evaluations, 0 failures**); the shared protocol (Supplementary S6) asks for the whole curve, so we then evaluated **all 806 retained checkpoints of the same 41 runs on the disjoint `test` split, 0 failures. The 202 shared (run, epoch) points agree to 0.0000 pp** — the same checkpoints evaluated twice, independently, on one machine and protocol: a free reproducibility check of the endpoint rule itself.

**One metric self-correction first.** Our first version used  $r(e) = \Delta T(e) / \Delta V(e)$ , producing values such as

**-860 % and +1031 %**: at late epochs  $V(e) \approx V(\text{final})$ , so the **denominator vanishes** —

**an ill-conditioned metric, not a finding.** We report only the non-dividing quantities

$\Delta V(e) = V(e) - V(\text{final})$ ,  $\Delta T(e) = T(e) - T(\text{final})$ , summarized by the slope  $\kappa$  of  $\Delta T$  on  $\Delta V$

**through the origin.**

Tables S21–S23 → Supplementary §S8.

**Slope  $\kappa$**  (how much of the movement on val transfers to test). On the **complete curve** —  **$n = 765$  (seed, epoch) points over the 41 runs —  $\kappa = 0.708$ ,  $R^2 0.937$** , with a run-clustered 95 % CI [**0.671, 0.745**] (sandwich SE 0.018, matched by a run-level bootstrap [0.672, 0.744]); the within-run estimator, which removes run-level differences entirely, gives **0.698**. The five-epoch sample gave **0.734** (not the independent unit — the epochs within a run are not), so **\*\*the completed sweep moves the estimate by  $-0.026$  with overlapping intervals: the conclusion is**

unchanged, and we now quote the complete-curve value as primary.\*\* The interval sits well inside

(0.5, 1.0), so the transfer rate is an estimate with a known width rather than a bare point value. Scripts:

work/x2\_kappa\_clustered\_20260918.py (reproduces the published 0.734 / 0.902 /  $n = 161$  exactly

before adding the interval) and work/x3b\_perepoch\_corrected\_20260918.py (complete curve). **\*\*The key point:  $\kappa$  is not a constant, and §8.6's 19 % does not contradict 0.734 — the two cover different**

regions of the curve\*\* (§13 puts both in one model: the realization rate is  $1 - WC/premium$ ).

- Across the curve (especially early) the movement is large —  $\Delta V$  reaches  **$-8.1$  pp** — and about **73 %**

transfers to test: **early-training progress is real progress**. §8.6's 19 % measures the

**selection-relevant sliver** near the peak, where  $\Delta V$  is of order  **$\pm 0.2$  pp** and  $\Delta T$  is

**directionless noise** (negative for the smoke2sf strategy arm).

- **Both are therefore true:** "most of the movement on val transfers"

( $\kappa = 0.734$ ) **and** "the little extra gained by selecting near the peak does not" (19 %) — the **selection premium is the residual of the transfer rate**.

- **And  $\kappa$  is not constant across arms either — in the direction the selection account predicts.**

On the complete curve, **baseline  $\kappa = 0.600$**  (95 % CI [0.581, 0.619],  $R^2 0.946$ ) and

**strategy  $\kappa = 0.767$**  ([0.735, 0.800],  $R^2 0.951$ ); resampling runs *within* each arm (never pooling

the two sets) puts the difference at **+0.168, 95 % CI [+0.130, +0.203]** (the five-epoch sample: 0.532 / 0.769 and **+0.237, [0.145, 0.323]**). **\*\*The arm that gains more from selecting on val (§5 Tier 1: the baseline arm) is the arm whose movement transfers less to test\*\*** — what a selection account requires, and an independent corroboration on data that already existed. **\*\*The completed curve narrows the arm difference by a third but keeps its sign and its interval away from zero\*\***: smaller than the sampled grid suggested, not larger.

**A second result — and a self-correction the complete curve forces.** With four sampled epochs we

wrote that **"the sign of  $\Delta_{\text{gap}}$  does not flip anywhere on the curve"**, and read it as non- $p$ -value support for §5.1 Tier 2. **\*\*On the complete curve that sentence is too strong for one of the two cells, and we narrow it:\*\*** from epoch 20 onward  $\Delta_{\text{gap}}$  is positive at every checkpoint in both cells,

and in smoke2sf it is negative over the first three — the grid's apparent uniformity was partly an artefact of where it sampled. The per-checkpoint values, the epochs that flip, and the check that reproduces the published values to the printed digit are in **Supplementary S11**.

**Three limitations:** ① only the two headline cells reach  $n = 10$ ; aitod20 has a single run (30 epochs

only, so it contributes 6 checkpoints where the others contribute 20 — hence 806, not 820); ② **closed: every retained checkpoint is now evaluated**, and the *stability* reading is narrower

than the sampled grid suggested, as stated above; ③  $\kappa$  is relative to each run's own final epoch and is **not** causal.

The complete 806-row evaluation is archived locally as  
xeval\_perepoch\_20260918/matrix\_perepoch.csv

with a SHA-256 list, verified against the pod copy after transfer.

## 9. Finding 6: the registered replication — hypothesis stated, criterion not met

**What was registered** (frozen 2026-09-13 01:37:27 UTC, **\*\*zero runs existing at freeze**

time, FROZEN-HASH md5 `6a7eee7b3e34b15ce5adcb14cf7ea36`). H1: on three unsaturated target

domains (100-epoch baseline mAP50-95 < 0.30), raising the peak learning rate from 0.001 to 0.005 during a 20 %-budget fine-tune (YOLOv12n [11] + shapeiou) yields a positive\*\* gain resolvable by ten-seed paired testing. "Replication succeeded" required  $\geq 2/3$  pairs

satisfying {gain  $\geq +0.30$  pp, paired  $p < 0.01$ ,  $\geq 8$  of 10 seeds same direction} and

surviving BH ( $q = 0.05$ ); otherwise report it as not replicated [28]. H2 (moderator): on

a saturated domain the same strategy's gain should be  $\approx 0$ . The ring design over

three aerial corpora makes source and target roles differ in every pair; 70 runs were frozen (three pairs  $\times$  2 arms  $\times$  10 seeds = 60, plus a saturated control of  $2 \times 5 = 10$ ), and the near-matched

control C was appended by amendment and later extended to ten seeds —  $2 \times 10 = 20$  further

runs — for 90 runs in total\*\* (70 registered + 20 appended); the count is taken from the run

directories themselves, ten seeds per arm in every cell except the saturated control, which has five.

**How the ten seeds are realized** (stated because the run metadata is misleading): the seed is carried by the shuffle order of the training files, not by the framework's seed argument, so

every run's args.yaml records the framework default `seed: 42` — expected, and not a sign

that the seeds coincide. What is verified: the per-seed results.csv are distinct in every cell

(20/20 runs in each registered pair, 10/10 in the saturated control, 20/20 for the appended

control), re-running one shuffle seed reproduces its run, and the same holds for the cells escalated

in §5.1. The mechanism, the framework version and the args.yaml detail are in **Supplementary S12**

(script work/check\_seed\_convention\_20260918.py).

**Result, at n = 10 per pair. T1-a (dota15→aitod) +0.147 pp** (CI [-0.036, +0.330],  $t = 1.816$ ,

$p = 0.1027$ , 6+/4-) — fails. **T1-b (aitod→visdrone) +0.168 pp** (CI [+0.083,

+0.253],  $t = 4.455$ ,  $p = 0.0016$ , 10+/0-, permutation  $p = 0.001953$ , the  $n = 10$  floor) —

passes direction and significance, fails the +0.30 pp magnitude criterion. **T1-c**

(visdrone→dota15) +3.727 pp (CI [+3.534, +3.920],  $t = 43.774$ ,  $p = 8.5 \times 10^{-12}$ , 10+/0-,

permutation  $p = 0.001953$ ) — **the only pair passing all three parts**. \*\*Significant after BH ( $q = 0.05$ ,  $m = 3$ ):  $2/3^{**}$  ( $2.5 \times 10^{-11}$  and  $0.0024$  survive; T1-a's  $0.1027$  does not).

**Passing the full registered criterion is a different count: 1/3**, because T1-b fails the magnitude part while passing significance. → **H1: not replicated** ( $1/3$  by the unit criterion,  $2/3$  after BH). We write "**criterion not met**", not "**hypothesis refuted**": T1-b is  $10/10$  in the same direction with its permutation  $p$  already at the attainable floor, yet its magnitude falls short — **direction stable, magnitude insufficient**, and that tension is itself the result.

**The saturated control** falls into **none of the frozen document's literal cases** (the registration requires recording such a case for human adjudication; we mark it a deviation): not " $\approx 0$  and non-significant" and not a positive gain, but **significantly negative** ( $-1.490$  pp,  $p = 0.0141$ ).

**The appended control C** (`dota` → `dota15`, class sets differing by one container-crane class) asks whether the gain depends on a label-space change: at  $n = 10$ , **+0.727 pp** (CI [ $+0.640$ ,  $+0.814$ ],  $t = 18.854$ ,  $p = 1.53 \times 10^{-8}$ , **10+/0-**). Its three-seed predecessor was  $+0.707$  ( $p = 0.0157$ ), **short of the threshold** — at  $n = 3$  the permutation floor is **0.25**, so it cannot reject there in principle; only ten seeds turn it into a verdict. Its YAML shares a directory between `val` and `test` and uses the project's reporting convention, the best endpoint — §10.

**How much of these  $\Delta$ s is the alias itself — val side.** In the four cells whose config points `val:` and `test:` at one directory, the metric watched during training **is** the reported metric, so each run's own log measures the selection premium directly:

$p = \max_e \text{mAP}_{50-95}(e) - \text{mAP}_{50-95}(\text{final epoch})$ . Paired across seeds within a cell,

$\bar{d} = \text{mean}[p_{\text{baseline}} - p_{\text{strategy}}]$  is the part of that cell's  $\Delta$  that is **\*\*differential**

selection opportunity\*\* rather than the intervention, so  $\Delta \approx \Delta_{\text{reported}} + \bar{d}$ .

The per-cell values of that term, with their intervals, are in **Supplementary S12** (audit script `work/x10_alias_audit_20260918.py`). In the same terms: T1-a **+0.198 pp**, T1-c **+3.941 pp**, control C **+1.133 pp**; T1-b is excluded by construction, since its YAML separates

the two keys, so its training-time metric is not the reported one and a val-side term cannot bound a two-term bias. **No verdict changes; one magnitude does**, and we state the smaller

number rather than leave the more striking one standing.

#### 9.4 What this section has to do with the paper's subject

**Four of the experiment's five cells rest on the `val`/`test` coincidence identified in §4** (three

registered pairs, the registered saturated control, the cell appended by amendment), so

**this verdict cannot be presented as a clean-protocol result:** it was obtained **\*\*under the protocol**

in force at registration time, **and §8's remediation ran on a** different batch of runs. One coincidence, stated without inferring

causation<sup>\*\*</sup>: the only cell with a separated split (T1-b) is the one whose magnitude falls short

(+0.168 < +0.30, despite 10/10 in the same direction), while the passing cell (T1-c) and the

significantly negative saturated control **are both on same-directory YAMLS**; with five

cells we **infer no mechanism**, yet it is an instance of "the reported split is not independent"

**landing on a concrete conclusion.** The registration and the later clean-protocol rerun

(r10\_p\_\*, 3 seeds) **agree in sign everywhere** (+3.61/+3.75/+3.38 versus

+3.74/+3.60/+3.80 on the largest cell) but are **\*\*not** the same batch and must not be pooled

or averaged<sup>\*\*</sup>; the frozen registration, per-pair tables, outcome-blind displacement measure

(including the published MAFA→mask = 7.1894) and verdict scripts are in

**Supplementary S24–S27.**

#### 10. Our own defects (disclosed, not hidden)

| #   | Defect                                                                                                                                                                                                                                                                                                                                                                                                                                     | Handling                                                                                                                                                                                                                                          |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --- | ---                                                                                                                                                                                                                                                                                                                                                                                                                                        | ---                                                                                                                                                                                                                                               |
| 1   | The clean-protocol results document <b>cites a ledger file that does not exist in the sense claimed, and the problem is worse than a wrong filename</b> : neither the cited <code>_r10_final_stats.txt</code> nor the file later identified as the "real log" ( <code>_r10_stats_0403.txt</code> ) contains <b>any</b> of that document's headline values (6 probes each, 0/6 hits) ⇒ <b>these numbers have no surviving stdout ledger</b> | <b>We do not point at a replacement file</b> (renaming would only manufacture a new false citation). Text citing those numbers notes <b>"no stdout ledger"</b> ; the p-values can be recomputed from the local <code>pvalue_table_local.py</code> |
| 2   | ~the dota15 cell of <code>selection_premium.py</code> was not reproducible~ → <b>fixed</b> (the root directory was a stale snapshot, and                                                                                                                                                                                                                                                                                                   | Stated in §5.3; <b>and it exposed a new disclosure</b> : M3 wrote "3 seeds" while only <b>2</b> runs carry a seed identity                                                                                                                        |

|   |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | the matcher used <code>prefix + '_s'</code> while the actual run names carry no seed suffix); after the fix it <b>reproduces M3's +1.12/+0.95/+0.93 value by value</b>                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3 | <code>'sio_b_results.csv'</code> has no written <b>endpoint convention</b> — <code>t2_mask2mende_base100_s42n</code> is recorded as final (67.12) in one archive and best (70.06) in another, i.e. "which checkpoint is recorded" was never specified                          | Stated in §7.3 with the consequences of each reading (significant versus not)                                                                                                                                                                                                                                                                                                                                                            |
| 4 | The two headline scripts <b>originally depended on a rented pod</b> ( <code>import paramiko</code> plus a plaintext password) — a paper about evaluation validity whose own analysis is not reproducible from the release                                                      | <b>Fixed:</b> <code>selection_premium_local_20260916.txt</code> and <code>r10_pvalue_table_20260916_local.md</code> now <b>read the local archive</b> ; the main text uses the local versions and lists the pod versions as cross-checks                                                                                                                                                                                                 |
| 5 | <b>No script computes the <math>\sigma</math> decomposition</b> (the three components come from three unrelated scripts); <b>no frozen effect-size rule file</b> ; <b>no script generates <code>'benchmark_split_audit.md'</code></b> (hand-written, 28 judgments as literals) | Each is labelled "manual / not archived", or a script is added                                                                                                                                                                                                                                                                                                                                                                           |
| 6 | The local archive <b>has no copy of the three-way carve directories</b> , and no copy of 6 of the 7 corpora                                                                                                                                                                    | <b>Partially repaired (2026-09-16, GPU run):</b> the per-epoch curves of the 41 three-way runs, <code>sio_b_results.csv</code> , and the $2 \times 2$ <b>evaluation matrix</b> (164 evaluations) are now local with a SHA-256 list ( <code>xeval_20260916/</code> ), so <b>§8's readings and §5.6's decomposition are locally recomputable</b> ; <b>what remains non-recomputable is retraining</b> , which needs the corpora themselves |
| 7 | The audit document has <b>internal inconsistencies</b> (nine items, found by our own census of it)                                                                                                                                                                             | Each adjudicated in <code>审计文档更正与裁定_20260916.md</code> : <b>3 turned out to be misreadings by the inventory, 4 are real defects, 1 is a convention risk</b>                                                                                                                                                                                                                                                                              |
| 8 | Our own corpora contain <code>val == test</code>                                                                                                                                                                                                                               | <b>That is this paper's argument</b> , not an embarrassment to be hidden                                                                                                                                                                                                                                                                                                                                                                 |
| 9 | <b>Our first per-epoch report claimed curve-wide stability from a four-point sample</b> , and reading the complete curve overturned it for one of the two cells (§8.7)                                                                                                         | <b>Corrected in place, with the retracted sentence quoted:</b> the complete sweep shows <code>smoke2sf</code> 's $\Delta$ gap negative at epochs 5/10/15 and positive from epoch 20 on, so the claim is <b>narrowed</b> rather than deleted, and the sampled grid's apparent uniformity is named as partly an artefact of where it sampled                                                                                               |

\*(The Chinese draft numbers these rows 1–3, 3, 4–7 with a duplicate "3"; renumbered here without

changing content.)\*

## 11. Recommendations

1. **Check the three keys before reporting:** do val and test point to the same directory or file list?

**At the tool level**, make "are val and test the same source?" a **hard pipeline check** —

§4.2: a tool produces this failure, a tool can block it.

2. **If you must select on the reported split:** say so ("the reported value is a **selected**

value") and **report a selection-free endpoint alongside it** (the final epoch, or the mean of the last

few). §5.1 shows this step **enlarges** the arm difference.

3. **Report the test convention:** below  $n = 6$  a permutation test has no resolution; at  $n = 3$  significance

comes from the paired t-test alone.

4. **Declare the unit:** any "how many benchmarks have a problem" number must state whether it is

release / protocol / yolo\_dist / reported.

5. **Select a plateau, not the single peak**, when you must select on the reported split: on 40 paired runs the val argmax captured **30 %** of the realizable gain and the best five val epochs captured **70 %** — at the cost of reporting a number **0.73 pp** below the peak, which is exactly the part of that peak that belongs to the split rather than to the model.

## 12. Limitations

- The audit reads **split definitions** (YAML, README, paper text), **no datasets downloaded**.

- For several benchmarks the "generic YOLO copy" is a **per-project** config, not a canonical artifact.

- Under the reported unit, **4 rows (AI-TOD, DIOR, Mendeley, D-Fire) are evidence-insufficient**; we do

not guess.

- Two evidentiary differences (pod versus local) are listed side by side, neither hidden.

- The census is a **literature/release survey**, weaker evidence than the **self-measurement** on our own

corpora (§3).

- The selection-premium decomposition (§5.6) covers **2 of 13 cells**; the other cells' runs are not on the

machine used — a scope we state rather than extrapolate.

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## Figure and table inventory

**Where the evidence lives.** This article keeps its \*\*claims, argument, four headline tables and its disclosure\*\*;

the remaining tables and all six figures are in the Supplementary Material, reproduced

**unchanged**, every number here still backed by an artifact pointer. Four tables are printed **here** too, because each carries a headline number to check:

| In this article | §S8 counterpart | What it settles                                                                            |
|-----------------|-----------------|--------------------------------------------------------------------------------------------|
| ---             | ---             | ---                                                                                        |
| <b>Table 1</b>  | Table S4        | <b>Tier 1</b> — the premium per arm and the paired arm difference, with t, p and 95 % CI   |
| <b>Table 2</b>  | Table S6        | <b>Tier 2</b> — the paired difference $\Delta$ gap by corpus, with the reading it supports |

|                |           |                                                                                                 |
|----------------|-----------|-------------------------------------------------------------------------------------------------|
| <b>Table 3</b> | Table S11 | the four counting units over the 19 audited benchmarks                                          |
| <b>Table 4</b> | Table S20 | the premium selected on val versus what that same selection realizes on the disjoint test split |

The rest are in §S8 and §S9, keyed to their main-text sections: **Table S1** (§2.1); \*\*Tables S2–S3, S5, S7–S10 (§3, §5, §5.1, §5.5, §5.6); Tables S12–S19\*\* (§6.1–§6.5, §7.2, §7.3); **Tables S21–S27** (§8.7, §9); the six figures:

| Figure         | Main-text section |
|----------------|-------------------|
| ---            | ---               |
| <b>Fig. S1</b> | §5.6              |
| <b>Fig. S2</b> | §8.6              |
| <b>Fig. S3</b> | §6.1              |
| <b>Fig. S4</b> | §5.5              |
| <b>Fig. S5</b> | §8.7              |
| <b>Fig. S6</b> | §13               |

§10's defect table is **kept in this article**, deliberately: disclosure belongs in the article. The Supplementary's own §0 lists how it is generated and verified, and its §S6 records the \*\*shared process protocols\*\* — the same set of protocols is used by this paper and its companion study — with their file hashes. Disclosure-type content — first-report attribution, methodological premises, gaps — stays \*\*in the article\*\*, per the venue's page convention.

### 13. Formalisation: what the four quantities are, and what they can and cannot identify

Sections 5–9 measure four quantities — the selection premium, the paired difference  $\Delta_{\text{gap}}$ , the transfer slope  $\kappa$ , and the realization rate. This section puts them in **one model** and states what each can and cannot identify. Proofs are in **Supplementary S11**; every proposition is anchored to numbers already measured in this paper.

**Setup.** Training is a randomised map (seed  $\omega$ , recipe  $\theta$ , checkpoint  $e$ )  $\mapsto$  weights; evaluation is a

fixed functional  $M_S(W)$  on a labelled split  $S$ ; write  $V(e) = M_{\{S_{val}\}}(W(e))$  and  $T(e) = M_{\{S_{test}\}}(W(e))$ ,

with  $\hat{e} = \operatorname{argmax}_e V(e)$  and  $e_f$  the final checkpoint.

**Proposition 1 (the raw gap is not identifiable).**  $V(\hat{e}) - T(\hat{e}) = [V(e_f) - T(e_f)] +$

$[(V(\hat{e}) - V(e_f)) - (T(\hat{e}) - T(e_f))]$ , i.e.  $G = \delta + o$  with  $\delta$  a split-difficulty term and  $o$  a

selection-optimism term. For any  $\delta'$  the pair  $(\delta', G - \delta')$  fits the observation equally well, so no

endpoint-pair estimator separates them [47, 48]. **The pairing assumption is what makes  $\Delta gap$  meaningful:**

with matched seeds and one split pair,  $E[\delta_{base} - \delta_{strat}] = 0$ , and only then does the arm difference

identify the difference in selection optimism. Under an aliasing configuration that assumption

*fails by construction* — the two arms'  $\delta$  are two maxima on the same split — which is why §9 cannot

be read as a clean-protocol result.

**Proposition 2 (exact three-component identity).** Differencing Proposition 1 across arms gives

$\Delta gap = \Delta \delta_f + \Delta prem_{val} - \Delta prem_{test}$ : the final-epoch arm difference on *val*, plus the two arms'

selection gains, minus their final-epoch arm difference on *test*. This is an identity, not a fit, and

it holds value by value on all 13 scanned cells and on both escalated cells at  $n = 10$  (§5.6).

**Proposition 3 (the sign of  $\Delta gap$  does not identify a mechanism).** Three measured facts carry this:

the components are of the same order (medians **1.07 / 0.54 / 0.93 pp**), so which one dominates

differs by corpus; **both signs occur** in the scanned cells (per-cell table: Table 2, Supplementary S6); every candidate mechanism we could

state is refuted by a

**sign mismatch** (M1/M2/M4/M5, Table S9). **What we do not claim** is that the sign is *fragile*: a perturbation analysis (numbers in **Supplementary S11**) leaves it stable under a  $\pm 10\%$  change of a single arm-difference term, so the honest statement is "the dominant term varies by corpus and the sign is a sum of same-order terms", not "the sign is noise".

**Proposition 4 (the realization rate is one minus the winner's-curse share).** With  $V(e) = v(e) + \varepsilon_e$

and  $T(e) = v(e) + \tau + \eta_e$ ,  $E[\text{premium\_test}] = E[\text{premium\_val}] - WC$  where  $WC := E[\varepsilon_{\hat{e}} - \varepsilon_{e_f}] \geq 0$ ,

so the realization rate is  $1 - WC / E[\text{premium\_val}] \leq 1$ , with equality only if selection is driven by signal alone [45, 46]. The inequality needs only that  $\hat{e}$  is chosen on val (which is the convention under

audit); the identity additionally needs the test noise not to enter the selection. **Two consequences, both tested on the archive** — the per-run inequality and the monotonicity in the val noise; per-run values, terciles and rank correlation: **Supplementary S11**. This is why §8.6's 19 % and §8.7's per-cell variation are the

same statement at different noise levels, not a contradiction. A third form — that the premium scales as  $\sigma\sqrt{2 \ln E}$  — was **tested and withdrawn**; the regression and its diagnostic are in **Supplementary S11**.

**Proposition 5 (Benjamini–Hochberg depends on the declared family).** The step-up thresholds are

$i/m \cdot q$ , so adding hypotheses with smaller p-values **raises** the thresholds behind them and the

rejection count is not monotone in the family. We are ourselves an instance: escalating two cells to

$n = 10$  changed the 13-cell count from **0 to 3** surviving cells (§5.1) [49, 50]. Declaring the counting unit

and the comparison family is therefore not a stylistic preference but a condition for the number to mean anything.

**Corollary (the premium is mostly shape, and the peak is the worst place to stand).** With the notation of Proposition 4,  $\text{premium}_r = \text{shape}(E) + v_r$  holds **exactly**, where  $\text{shape}(E)$  is the gap

between the mean curve's peak and its final epoch. Measured over 40 runs at  $E = 50/100/200/400$ , shape is

**71 %/91 %/88 %/9 %** of the premium. The four-budget series is **not monotone**: the premium peaks at  $E = 200$

(+0.971 pp) and **falls** at  $E = 400$  (+0.907 pp), so the pre-declared monotone-rise criterion is **\*\*not met; and  $\text{premium}(400)/\text{premium}(100) = 1.329$  lies inside\*\*** the pre-declared band  $[0.86, 1.43]$

around the  $\sigma\sqrt{2 \ln E}$  prediction 1.141, so that scaling is **not rejected** — reading the level as

shape-driven rather than jitter-driven is therefore a *candidate*, not a result. **The four early-stopped runs decide the outcome** — one restriction (dropping them) flips both

criteria — and the clean test, the same budgets with early stopping disabled, was not run; we report

the pre-declared verdict as **not met / scaling not rejected**, with this dependence stated rather than resolved. The per-budget detail is in **Supplementary S11**. Two consequences follow. (i) A low realization rate is expected: most of the val premium is the split's own shape, not transferable gain. (ii) The val **\*\*argmax** is a poor checkpoint to carry over\*\*, and the reason is stated as a candidate rather than a claim: the measured test for it is in **Supplementary S11**. On 40 paired runs (selection on val, readout on a disjoint split) the argmax captured **30 %** of the realizable gain and the two peaks agreed in only **17.5 %** of runs, whereas **averaging the best five 'val' epochs captured 70 %**. Derivations and the full tables are in **Supplementary S11**.

### 13.1 Five tests on GPU machines, each with its criterion written down first

The propositions above are testable, so we ran them on rented GPUs with the **\*\*criterion** fixed before the

runs\*\*. Two of the five outcomes are negative and one design had to be redone — reported as they came out.

The six experiments, their **pre-declared criteria** and how each came out are tabulated in **Supplementary S31–S33**: G1 **void** (the two splits differ by 14.7 pp in difficulty on the same checkpoints, so its  $-0.582$  pp mixes the effect with a level effect); G1'' **met**; G1' **not met** (WC =  $-0.069$  pp,  $t = -0.23$ ,  $p = 0.83$ ,  $n = 10$  — no inflation detectable); G2 **\*\*direction met**, magnitude not met; G3 **met**; G4 **met** (WC =  $+5.21$  pp\*\*, CI [ $+4.82$ ,  $+5.66$ ], 20/20 runs positive). Where the outcome is negative it is reported as negative.

The four criteria that were met and the two that were not carry one message together: **\*\*the selection**

bias moves with the absolute amount of selection noise and with the number of selection opportunities,

but not with whether the two splits are the same file.\*\* Two consequences do not follow from the numbers

alone: the configuration-level fact of §4 stands — the reported number *is* the selected number — while

the stronger reading, that aliasing *adds* a measurable amount on top of it, does not. Proposition 4's WC

is best read as the winner's-curse share of the selected gain, a quantity set by the noise. The per-comparison reasoning and the sources are in **Supplementary S31–S33**.

## Declarations

- **Data availability.** All quantitative claims are backed by local artifacts listed in the Supplementary

Material; the third-party training logs are **not redistributed** (hashes and source pointers only), and

5 of their sources state no licence.

- **Competing interests.** None declared.

- **One quantity, two spellings.** The  $n = 10$  attainable floor is **exactly**  $2/2^{10} = 0.001953$ ; the tables use the exact value while the abstract writes **0.0020** in the same place. Both denote the same floor, so any mechanical consistency check must compare **numerically** ( $|\Delta| < 5 \times 10^{-5}$ ) rather than as strings.

- **CRedit author statement.** \*[To be completed with the final author list, using the 14 Elsevier roles: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.]\*

- **Reproducibility.** Every number in this paper carries a pointer to a local artifact; 291 audit anchors

mechanically guard the claims, and six checkers (`audit_anchors_fupaper.py`, `check_crosscite.py`,

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